ABSTRACT

The proliferation of multi-tenant Enterprise Resource Planning (ERP) systems has democratized access to embedded Machine Learning (ML) services for Small and Medium-sized Enterprises (SMEs). Modern ERP platforms ship with predictive modules for late-payment risk, lead scoring, and employee attrition, and increasingly augment these models with post-hoc Explainable AI (XAI) techniques such as SHAP and LIME. While the data-rich enterprise regime is well served by this combination, the long tail of cold-start tenants those with only tens or hundreds of historical records reveals a previously under-examined failure mode: hallucinated explanations. SHAP and LIME continue to produce confident, well-formatted attributions even when the underlying classifier is statistically equivalent to random guessing, exposing SME users to authoritative-looking but epistemically empty recommendations. This thesis quantifies the phenomenon through a controlled sparsity experiment across five simulated tenants (50,000; 5,000; 1,000; 200; 50 rows) drawn from the Brazilian Olist e-commerce corpus, operationalised as a Late Delivery Risk prediction task (1[delivered_customer_date > estimated_delivery_date]) that serves as a structural ERP-style operational-risk substrate generalising to the studied system's late-payment module, and trained under a strict leakage-avoidance regime in which only order-time observables are admitted into the feature matrix. Classical predictive metrics fail to track the regime change: AUC-ROC remains uniformly modest across the data-rich, medium, sparse, and very-sparse tiers (0.6556, 0.6249, 0.6094, 0.6126) and rises anomalously to 0.8889 at the cold-start tier as a small-sample artefact. Baseline rank agreement of SHAP explanations against the data-rich reference tenant (a reference-aligned explanation-stability measure, not a ground-truth fidelity claim), quantified by the Spearman rank correlation of global SHAP importances, drops monotonically from $\rho = 1.000$ to $\rho = 0.4485$, crossing below the $\rho = 0.70$ reliability threshold at the 1,000-row mark ($\rho_C = 0.6429$). Explanation volatility, measured by the variance of absolute SHAP values, expands by more than two orders of magnitude from 0.0224 at the data-

rich tier to 5.0392 at the cold-start tier. The qualitative cliff is replicated under the same pipeline on two further ERP-native tasks (IBM HR Analytics Employee Attrition; synthetic CRM Lead Scoring), supporting the external validity of the architectural finding. Building on these empirical findings, the thesis proposes and implements a Maturity-Aware AI Routing pattern in a Django 4.2 multi-tenant ERP: a tier-logic gate (LatePaymentPredictor.predict_or_heuristic) that suppresses local predictions for tenants below the 1,000-invoice threshold and substitutes platform-wide global heuristics derived from the data-rich reference. The proposed architecture provides a concrete, auditable systems-level control against hallucinated explanations and offers SaaS vendors a scalable cold-start fallback consistent with emerging AI-accountability regulation.

Keywords: Explainable AI, SHAP, LIME, multi-tenant SaaS, ERP, cold-start, data sparsity, Django, reliability cliff.